

Sanskrit Document Retrieval-Augmented Generation (RAG) System

1. Introduction

This project focuses on building a Retrieval-Augmented Generation (RAG) system for Sanskrit documents. The goal is to accurately retrieve relevant Sanskrit text passages from documents and generate correct, context-aware answers to user queries using Large Language Models (LLMs).

2. Problem Statement

Traditional language models may hallucinate answers when asked questions from specialized or low-resource languages like Sanskrit. The challenge is to ensure factual correctness by grounding the model's responses in the original document content.

3. Objectives

- Build a document ingestion pipeline for Sanskrit PDFs/DOCX files.
- Create vector embeddings for efficient semantic search.
- Retrieve the most relevant document chunks for a query.
- Generate accurate answers strictly based on retrieved context.

4. Dataset Description

The dataset consists of Sanskrit documents in PDF and DOCX formats. These documents contain classical Sanskrit text related to historical, literary, or philosophical content.

5. Methodology

The system follows a standard RAG pipeline: document loading, text splitting, embedding generation, vector storage, similarity-based retrieval, and answer generation using an LLM.

6. System Architecture

1. Document Loader: Extracts text from PDF/DOCX files.
2. Text Splitter: Breaks large documents into manageable chunks.
3. Embedding Model: Converts text chunks into numerical vectors.
4. Vector Database (FAISS): Stores embeddings for fast similarity search.
5. Retriever: Fetches relevant chunks based on the user query.

6. LLM: Generates final answers using retrieved context only.

7. Tools and Technologies Used

- Python
- LangChain
- FAISS Vector Store
- HuggingFace / OpenAI Embeddings
- PyPDF2, python-docx

8. Evaluation Strategy

The system is evaluated using manually designed questions from the document. Generated answers are checked for correctness, relevance, and faithfulness to the source text.

9. Results and Observations

The RAG system successfully retrieves relevant Sanskrit passages and produces more accurate and grounded answers compared to a standalone language model. Hallucinations are significantly reduced.

10. Limitations

- Quality depends on OCR/text extraction accuracy.
- Sanskrit language complexity can affect embeddings.
- Limited evaluation dataset.

11. Future Enhancements

- Use Sanskrit-specific embedding models.
- Add multilingual query support.
- Improve chunking strategies.
- Deploy as a web application.

12. Conclusion

This project demonstrates that Retrieval-Augmented Generation is an effective approach for question answering over Sanskrit documents. It ensures factual accuracy and makes LLMs more reliable for low-resource and classical languages.